Engineering Mechanics 1 · Unit_03 · UNIT 3

DRAFT — UE26ME142A · Unit_03 · UNIT 3 · CLASS 35 Centroid Numerical

Standalone video lesson · authored study notes with source citations

Hugging Face PESU_Materials source tree, official units.json catalogue, transcript sidecars, slide text/PDF/PPT sources and the selected First_Year_8enterprise archive; see the provenance datasets in the release ZIP. Generated 18 October 2026.

· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

Contents

1. 1 · CLASS 35 Centroid Numerical
2. Appendix A · Symbol glossary
3. Appendix B · Sources, coverage and gaps
4. Appendix C · Revision and retrieval practice

M1

CLASS 35 Centroid Numerical

1 CLASS 35 Centroid Numerical

Study focus: CLASS 35 Centroid_ Numerical. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000058 · EV0000059

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

How to locate the centroid of the given figure.

Key details

- This is the question 5/55.
- Locate the y coordinate of the centroid of the shaded area.
- Look at this figure.
- In the sector, they have removed this triangle, and we have been asked to locate the centroid of only this shaded area, and that to only the y-coordinate.
- They have not asked about the x-coordinate.
- If they ask about x coordinate, then where will be the centroid?
- Look at the figure.
- Due to symmetry, the centroid will be on the y axis so x coordinate of the centroid will be 0.
- Since they have not asked, it's not necessary to mention about x bar.
- We need to find only the value of y bar.
- Next, how to get this shaded area?
- To get the shaded area, we have to use the method of approximation.
- We have to divide this into elemental areas.
- We can divide this as a sector and remove this triangle from the, then you will get this shaded area.
- There is a sector, the inscribed angle will be 90 degrees, nothing but pi by 2.
- Then radius is A.
- That's about sector.
- Then come to this triangle.
- Here the height of the triangle will be H.

- Then what will be this base?
- What will be this base of the triangle that we don't know?
- I am going to call the basis B since B is the term which I introduced in the.
- Question, I should know the value of B in the form of H.
- Then how to express B in the form of H.
- If this is 45 degrees, this angle will also be 45.
- Then due to symmetry, if total is B then this will be B by 2.
- Tan 45 is equals to b by 2 by H.
- Sorry, we will get the value of B.
- B is equals to 2 into H.
- First one, let it be a sector of radius A, and the angle alpha will be here will be equal to this will be alpha.
- In the sector derivation total angle was 2 alpha.
- Here also 2 alpha is pi by 2, that is 90 degree.
- Next triangle, second one.
- Base of the triangle will be 2 into h and the height of the triangle is h itself.
- And we know the formula for y bar, summation of a i yi divided by total area.
- area of the first elemental element.
- What is the formula to find area of a sector?
- R square into alpha.
- Remember alpha should be in terms of radians.
- This A1 area of this sector.
- Then we should do the value of Y1.
- What is the meaning of Y1?
- This sector will have a centroid that is C1, then the Y coordinate of that C1 will be Y1.
- Since the sector is symmetrical about y-axis, the centroid of the sector will also be on the y-axis.
- Then this is y1.
- How to find this y1?
- To find the Y1, we know the distance of the centroid of this sector with respect to its center and the axis of symmetry is 2 R sin alpha by 3 alpha.
- This is the formula standard formula.
- Next is triangle.
- Since this triangle is removed or it's not shaded, area should be taken as negative.
- What is base.
- Two times the dark h and the height is h itself.
- Substitute here, you will get the value of A2S minus h square.
- After that, we should know y2.
- What is y2?

Example or application from the lecture

- Here, they have given a sector.
- Then, before getting into the method of approximation, let's know about the dimensions of the sector as well as the triangle.

- For that I am going to consider only this part of the triangle, that is right angle triangle.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000058 · EV0000059 · EV0000060 · +1 source units in pack.json

Core explanation · 05:00–09:59

This triangle will have a centroid that will be C2.

Key details

- The y coordinate of C2 is Y2.
- Look at this axis, x-axis is passing through the ax of the triangle.
- That will be 2H by 3.
- The general formula is a1 y1 plus a2 y2 by a1 plus a2.
- But since the value of A2 is negative, substitute that as it is here.
- That's why you will get the second term as negative here as well as in the denominator also.
- Once you simplify you will get the value of y bar.
- This is the value of y bar.
- That's the point where the entire area of the shaded region is assumed to be concentrated.
- Determine the coordinates of the centroid of the shaded area.
- This is the shaded area.
- We have been asked to locate the centrate of only shaded area.
- Then how to locate the centr of only shaded area?
- Here also we have to use the method of approximation, divide this into elemental areas, then use the formula of x bar and y bar, find out x bar and y bar.
- How to divide this into elemental areas.
- First if you draw a vertical line passing through this point, this will be triangle, then.
- This is semicircle, so the remaining will be rectangle here, then semicircle.
- In that we have to deduct this rectangle, then semicircle of radius 30.
- Only then we will get the shaded area.
- So this is the first one triangle.
- Second one is this rectangle.
- And the third one is this bigger semicircle of radius 40.
- In that we have to deduct fourth one, that is semicircle of radius 30.
- Fifth one is also deduction, that is a rectangle of width 50 and the height.
- Then only we will get this shaded area.
- We know the formula x bar and y bar a1 y1 plus a2 y2, so on divided by total area.
- First one is the triangle of base 40 and the height.
- Height will be the this that will be diameter of this bigger semicircle A T.
- That is area of the triangle.
- Then we need to find X1 Y1.
- X1, Y1 are the coordinates of the centroid of this triangle.
- The centroid of the triangle will be somewhere here, that will be C1.
- The x coordinate is this, that is x1.
- Then how to find this x1?

- We know the standard formula, this x1 will be equals to with respect to this side, it will be two times the of base by three.
- Next, we should know the value of y1.
- With respect to base of the triangle, y coordinate will be h by 3, here h is this AT.
- We are done with the first elemental area.
- The width of the rectangle will be how much?
- Total will be 140 minus 40, because this 40 is included in the triangle.
- Next we should know X2 Y2.
- Ah, this rectangle will have a centrite somewhere here at the center, C2, x coordinate of this, that is the distance from y axis to c2, that is x2.
- This is x2.
- Then how to find this x2?
- From the derivation we know from ear to ear distance will be b by 2.
- But x2 is from y axis, so we have to add this 40.

Example or application from the lecture

- Let's move on to the next problem 5 slash 58.
- Let's start with the first one.
- Okay, let's start with the first one.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000062 · EV0000063 · EV0000064 · +1 source units in pack.json

Core explanation · 10:00–14:59

Then y2 is the y coordinate of this c2, this distance, that will be equal to total 80, 80 by 2 will be 40.

Key details

- We are done with the second lamina.
- Come to the third elemental area.
- The third area is semicircle of radius forty.
- I R square by 2 R is 40.
- You will get the value of area of this bigger semicircle.
- Next, this bigger semicircle will have a central.
- Where will be that centroid?
- On the axis of symmetry, somewhere here.
- If you draw a horizontal line passing through the center, this will be the axis of symmetry.
- Then what is X3?
- X3 is this distance.
- From y axis to C3.
- Plus what is this distance?
- That will be 4 r by 3 pi.
- Since it's a we are talking about bigger semicircle, radius is 40.
- Next y 3.

- Y3 is y coordinate of this C3 that will be equals to the di radius itself that will be 40.
- Next is the deduction part, that is fourth one, we have to deduct this smaller semicircle of radius 30.
- Next, where will be the centroid of this smaller semicircle?
- Again, that will be on the axis of symmetry somewhere here, C4.
- Then what is X4?
- Ear to ear distances X4.
- Then how to find the 64?
- From year to year we know 140 plus center to centroid will be 4r by 3 pi.
- Since C4 is in the first quadrant, it will be positive.
- 140 plus 4 R by 3 pi ERR is 30, you will get the value of X4.
- Then what is Y4?
- Y4 will be the diameter of sorry radius of this circle that will be 40.
- Next, the last is the deduction of this rectangle.
- What is b here?
- 50 and what is the height of this rectangle?
- You'll get it as minus 2000.
- Then we should know x phi and y phi.
- X phi and y phi are the.
- Coordinates of the centroid of this C phi.
- Then where will be what is this x phi?
- This distance is x phi.
- Then how to find that?
- That is 40 from here to here 40 plus 20 plus what is this?
- 50 by 2.
- We will get it as 85.
- Then we should know y2, sorry, y phi, y phi will be equals to.
- This 80 by 240.
- We got the value of all the terms in the equation.
- In this formula, so substitute here a1 x1 then plus a2 x2 plus a three x three plus.
- Then you should open the bracket, since it is minus it should be.
- Minus 450 pi into plus of 152.7.
- Similar next plus of minus 2000 into 85 divided by total area will be a one plus a2 plus a three.
- Since it is minus it will be minus four fifty pi 450 into pi, then minus two thousand.
- Once you simplify it, get the value of x bar.
- A1 Y1 A2 Y2 A3 Y 3 A4 Y 4 A5 Y 5 You'll get the value of Y bar.
- The Y bar is 37.55.
- This is the point 88.67, 37.55 will give us the coordinate of the centroid of the shaded area.
- That's the point where we will assume the entire area is concentrated.
- That's about today's session.
- If you come across any doubts, you can post your queries in the PESO forum.

Example or application from the lecture

- Then what will be this total distance that will be equals to from year to year 140 is given here.
- try to solve the problems given in the assignment as well as in the cushion bank.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000066 · EV0000067 · EV0000068 · +1 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
CLASS 35 Centroid_ Numerical transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/036_CLASS_35_Centroid__Numerical__6bf222208dbdadf5.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
CLASS 35 Centroid_ Numerical transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/036_CLASS_35_Centroid__Numerical__6bf222208dbdadf5.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
CLASS 35 Centroid_ Numerical transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/036_CLASS_35_Centroid__Numerical__6bf222208dbdadf5.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9991; only the JSON segmentation supplies the displayed timestamps.
CLASS 35 Centroid_ Numerical transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/036_CLASS_35_Centroid__Numerical__6bf222208dbdadf5.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. 15 rendered pages in 033_Centroid__9f0e55503970de0e.slides.md have no readable `.slides.md` text; their original page images are included.
2. 2 rendered pages in 033_UE26ME142A_Unit_3_Class_33_Lecture_PPT__ce948fc3cd6a59bf.slides.md have no readable `.slides.md` text; their original page images are included.
3. 3 rendered pages in 039_UE24CV141A__Unit_3_Beams__9fc34bbd5ca57988.slides.md have no readable `.slides.md` text; their original page images are included.
4. 3 rendered pages in 039_UE25CV141A_Unit_3_Class_48_Lecture_PPT__4d28f283532e284e.slides.md have no readable `.slides.md` text; their original page images are included.
5. 3 rendered pages in 042_UE25CV141A_Unit_3_Class_53_Lecture_PPT__c20af5d61d14aaf0.slides.md have no readable `.slides.md` text; their original page images are included.
6. 5 rendered pages in 043_UE25CV141A_Unit_3_Class_54_Lecture_PPT__c7db1f404daea47d.slides.md have no readable `.slides.md` text; their original page images are included.
7. 6 rendered pages in 044_UE25CV141A_Unit_3_Class_55_Lecture_PPT__741ad989a632bc04.slides.md have no readable `.slides.md` text; their original page images are included.
8. 4 rendered pages in 044_UE26ME142A_Unit_3_Class_44.1_Lecture_PPT__068ee897e4f8117e.slides.md have no readable `.slides.md` text; their original page images are included.
9. No historical archive question page met the unit-match candidate threshold; all subject-level and ambiguous pages remain in datasets/question-archive.json.
10. 6 video-to-slide associations were not promoted to a match. Candidate rankings are listed in the transcript corpus and video-slide-matches.csv; the slide decks appear separately.
11. 1 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.
12. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
13. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
14. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
15. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00014	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0043. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0044 to s0064. Sidecar variants compare...
SRC00015	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0043. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0044 to s0064. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00016	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0043. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0044 to s0064. Sidecar variants compare...
SRC00017	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0043. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0044 to s0064. Sidecar variants compare...

C Revision and retrieval practice